

Devin Joseph Johnson

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EDUCATION

- PhD – Statistical Science, Duke University (2021-Present)
 - Committee: Eric B. Laber, Jason Q. Xu, Simon Mak
 - Areas of Study – Reinforcement Learning, Computational Statistics, Active Learning and Optimization
- Master of Science – Statistics, Brigham Young University (2021)
 - Thesis: Search Algorithms and Loss Functions for Bayesian Clustering
 - Committee: David B. Dahl (chair), Garritt L. Page, Robert A. Richardson
 - *Summa Cum Laude*, (GPA: 4.0)
- Bachelor of Science – Statistics, Brigham Young University (2021)
 - Minors: Mathematics, Computer Science
 - *Summa Cum Laude*, (GPA: 4.0)
- Duluth East High School (2015)
 - *Summa Cum Laude*, valedictorian (GPA: 4.0)

PROFESSIONAL EXPERIENCE

- Undergraduate Research Assistant – Brigham Young University (May 2018 – Aug. 2019)
- Graduate Research Assistant – Brigham Young University (Aug. 2019 – Aug. 2021)
- Graduate Teaching Assistant – Duke University (Aug. 2021 – May 2024)
- Decision Science Graduate Intern – Disney (June 2024 – Present)
 - Worked with large and complicated datasets for Disney Cruise Line and Disney Publishing, analyzing these data to pull out key insights to inform decisions
 - Presented findings in a clear manner to key decision makers and an audience with varying degree of technical background

PUBLICATIONS

- D.B. Dahl, **D.J. Johnson**, and R.J. Andros. “Comparison and Bayesian Estimation of Feature Allocations” *Journal of Computational and Graphical Statistics*.
<https://doi.org/10.1080/10618600.2023.2204136>
- D.B. Dahl, **D.J. Johnson**, and P. Müller. “Search Algorithms and Loss Functions for Bayesian Clustering” *Journal of Computational and Graphical Statistics*.
<https://doi.org/10.1080/10618600.2022.2069779>

SOFTWARE

- **salso** – R package for SALSO algorithm for clustering comparison and estimation. Co-author with David B. Dahl. **62K+** downloads.
- **fangs** – R package for FANGS algorithm for feature allocation comparison and estimation. Co-author with David B. Dahl. **27K+** downloads.

PRESENTATIONS

- D.B. Dahl, **D.J. Johnson (presenter)**, and R.J. Andros. “Comparison and Bayesian Estimation of Feature Allocations”, Graduate Student Research Seminar Series, Department of Statistical Science, Duke University, Durham, NC, September 2023.

- D.B. Dahl, **D.J. Johnson (presenter)**, and P. Müller. “Search Algorithms and Loss Functions for Bayesian Clustering” Student Research Conference, College of Physical and Mathematical Sciences, Brigham Young University, Provo, UT, February 2021.
- D.B. Dahl, **D.J. Johnson (presenter)**, and P. Müller. “Search Algorithms and Loss Functions for Bayesian Clustering” Student Research Conference, College of Physical and Mathematical Sciences, Brigham Young University, Provo, UT, February 2020.
- D.B. Dahl (presenter), **D.J. Johnson**, and P. Müller. “Summarizing Distributions of Latent Structures” Bayes Comp 2020, Gainesville, FL, January 2020.
- D.B. Dahl, **D.J. Johnson (presenter)**. “Sequentially-Allocated Latent Structure Optimization (SALSO) algorithm” Student Research Conference, College of Physical and Mathematical Sciences, Brigham Young University, Provo, UT, March 2019.
- D.B. Dahl (presenter), **D.J. Johnson**, and P. Müller. “Summarizing Distributions of Latent Structures” Pontificia Católica de Chile, Santiago, Chile, January 2019.
- D.B. Dahl (presenter), **D.J. Johnson**, and P. Müller. “Summarizing Distributions of Latent Structures” ISBA World Meeting, Edinburgh, UK, June 2018.

AWARDS

- Thomas S. Monson Presidential Scholarship, Brigham Young University (2017-2020)
- Dean’s List, College of Physical and Mathematical Sciences, Brigham Young University (Fall 2017, ‘18, ‘19, Winter 2018, ‘19, ‘20, Spring/Summer 2018, ‘19)
- Robert A. and Sophia B. Whiteside Scholarship, Duluth, MN (2017-2021)
- First Place, Domo/Weave data science team hackathon (February 2020)
- BYU Statistics Undergraduate Mentoring Spotlight (Spring 2019)
- National Merit Scholar (2015)

SKILLS

- Programming languages: Expert in R, competent in Python, C++, Java
- Proficiencies: LaTeX, Beamer, Linux command line, Git version control, MS Office
- Languages: English (*native*), Portuguese (*fluent*)

SELECTED COURSEWORK

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| • Data Structures | • Analysis of Correlated Data |
| • Applied R Programming | • Generalized Linear Models |
| • Analysis of Variance | • Bayesian Statistical Modeling |
| • Advanced Programming Concepts | • Probability and Measure Theory |
| • Probability and Inference 1&2 | • Statistics Case Studies |
| • Algorithm Design and Analysis | • Statistical Inference |
| • Software Design and Testing | • Probability and Statistical Models |
| • Technical Communication | • Bayesian Causal Inference |
| • Theory of Analysis | • Modern Causal Inference |
| • Linear Models | • Bayesian Nonparametrics |
| • Probability Theory and Math Stat 1&2 | • Reinforcement Learning |
| • Statistical Learning and Data Mining | • Computational Statistics |
| • Mixed Model Methods | • Active Learning and Optimization |
| • Multivariate Statistical Methods | • Probabilistic Machine Learning |
| • Deep Learning | • Advanced Statistical Computing |